THE SHEAFIFICATION OF A K-FLAT COMPLEX OF MODULES OVER AN AFFINE RING IS K-FLAT

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Allowed standard prerequisites (editorial).

- Localization of a K-flat complex of modules is K-flat over the localized ring. invoked at proof lines 551-553. Gives K-flatness of the stalk complexes.
- A complex of sheaves is K-flat iff all its stalk complexes are K-flat. invoked at proof lines 554-558. Concludes.

Source reference. Original lemma 06Y1: sources/stacks-chow/tags.txt, source lines 8870–8870. Cited at proof lines 551-552 for the K-flatness of the stalk complexes; the sheaf-level conclusion is inside the counted span.

Source reference. Original lemma 06YB: sources/stacks-chow/tags.txt, source lines 8880–8880. Cited at proof lines 555-555 to pass from stalks to the sheaf; the conclusion is inside the counted span.

Source reference. Original lemma 01HV: sources/stacks-chow/tags.txt, source lines 1956–1956. Cited at proof lines 557-557 for the identification of the stalks; the conclusion is inside the counted span.

Lemma 0.2. *Let $X = \mathrm{Spec}(A)$ be an affine scheme. If $K^\bullet$ is a K-flat complex of A-modules, then $\widetilde{K^\bullet}$ is a K-flat complex of $\mathcal{O}_X$-modules.*

Proof. By More on Algebra, Lemma 06Y1 we see that $K^\bullet \otimes_A A_\mathfrak{p}$ is a K-flat complex of $A_\mathfrak{p}$-modules for every $\mathfrak{p} \in \mathrm{Spec}(A)$. Hence we conclude from Cohomology, Lemma 06YB (and Schemes, Lemma 01HV) that $\widetilde{K^\bullet}$ is K-flat. □

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